

Graphs and Their Relation to d -Edge Density

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Outline

- 1 Introduction
 - Graphs
- 2 Important Definitions
 - Edge Blankets
 - Internal Edges
 - d -edge density
- 3 Results
 - Proposition #1
 - Proposition #2
 - Proposition #3
 - Flower Theorem
 - Theorem case 1 and 2
 - Theorem case 3
 - Cycle Graphs
 - Complete Graphs
 - An application to commutative algebra
- 4 Thank you!

Introduction

Graphs

A part of our summer has been focused on a particular sub-topic of mathematics called Graph Theory, and instead of dealing with graphs in the xy -plane, this research focused on the combinatorial object consisting of edges and vertices. They can have finite combinations of vertices and edges.

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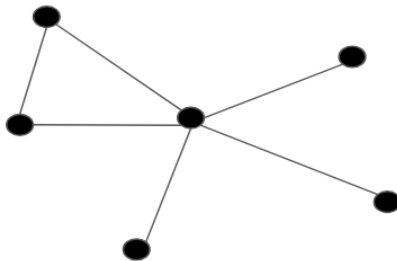


Figure 1: An example of a graph.

Graphs

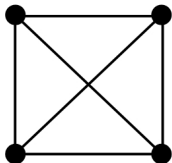


Figure 2: The graph K_4 .

Graphs

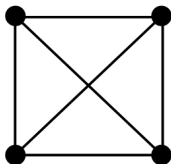


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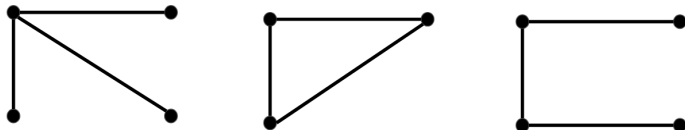


Figure 3: Subgraphs of K_4 with 3 edges.

Important Definitions

Edge Blankets

Definition

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. A subset of edges $J \subseteq E(G)$ is an **edge blanket** if for all $e \in E(G)$, we have $V(e) \cap V(J) \neq \emptyset$.

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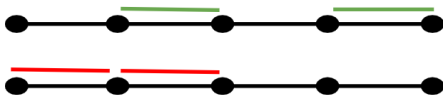


Figure 4: Example of a L_5 Graph, the green lines show a subgraph that has the properties to form an Edge Blanket while the red lines show a counterexample.

Internal Edges

Definition

Let G be a graph with the set of vertices $V(G)$ and the set of edges $E(G)$. A subset of edges $J \subseteq E(G)$ has an **internal edge** if there exists an $e \in J$ that is internal, i.e., $V(e) \subseteq V(J \setminus \{e\})$.

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Figure 5: Example of a L_5 graph, the central green line represents an internal edge. The two red lines are a subgraph and are not internal edges.

D-Edge Density

Definition

Let d be a positive integer. A graph G is **d -edge dense**, if for all subsets $J \subseteq E(G)$ of size d , J is an edge blanket or it has an internal edge.

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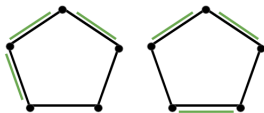


Figure 6: C_5 Subgraphs of 3 Edges



Figure 7: C_5 is Not 1- or 2- Edge Dense

Results

Proposition #1

If G is a graph with n -many edges, then G is n -edge dense.

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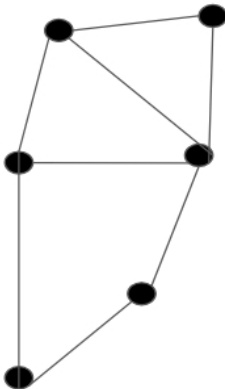


Figure 8: Example of a graph with 8 edges.

Proposition #2

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Proof.

For something to be d -edge dense, then all possible combinations of subgraphs that have d amount of edges are either edge blankets or have internal edges. Looking for $(d + 1)$ is unnecessary since then you would be adding edges to possible subgraphs that already are edge blankets or have an internal edge. □

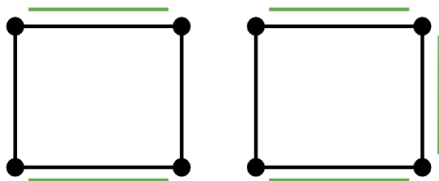


Figure 9: A subgraph of C_4 with 3 edges built from a subgraph of C_4 with 2 edges.

Criteria

Question: What is the smallest edge density that a graph can have?

To show that something has a minimal edge density of d , we have to use proof techniques to show:

- It is not $(d - 1)$ -edge dense, by finding a subgraph counterexample
- It is d -edge dense, either directly or through induction.

Proposition #3

Definition

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. G is a **star graph** if there exists a $v \in V(G)$ such that for all $e \in E(G)$, we have $V(e) \cap \{v\} \neq \emptyset$.

Proposition

For any graph G that is a star graph, its minimal d -edge density will be 1.

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For any graph G that is a star graph, its minimal d -edge density will be 1.

Proof.

We can take any set of edges $E(G)$ that includes the center vertex with any other vertex on the graph to form an edge. This particular edge ends up touching every possible edge set resulting in an edge blanket. □

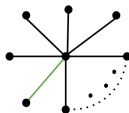


Figure 10: S_n Star Graph for any n edges

Theorem

Definition

A **flower** is a graph where all the petals share a vertex in the center of the graph. A petal consists of three edges and three vertices, including the shared vertex. (e.g. a triangle). Two edges of a petal are connected to the center vertex and the third edge is connected between the two non-center vertices.

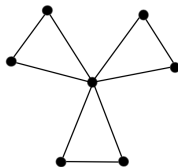


Figure 11: 3 petal flower, written F_3 .

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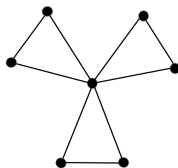


Figure 11: 3 petal flower, written F_3 .

Theorem

The minimum edge density of every flower graph, F_k , is $2k - 1$, for k the number of petals. All flowers have an odd minimum edge density.

Proof of Theorem

Proof.

We will show F_k 's edge density cannot be $2k - 2$. In this figure, it provides the counterexamples as petals go up. All counterexample subgraphs with an even number of edges are not d -edge dense because there's an edge not being touched (indicated by the dotted line). But adding an edge anywhere will make an edge blanket or internal edge.

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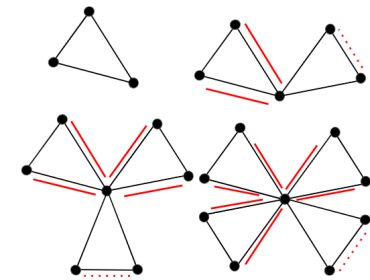


Figure 12: Flower graph counterexamples.

Proof.

Now we will use induction.

Base Case: $F_1 = C_3$.

Inductive Step: For us to prove the edge densities of flower graphs, we have to assume F_k has a minimum edge densities of $2k - 1$ for some $k > 1$. We will show that F_{k+1} has a minimum edge density of $2(k + 1) - 1 = 2k + 1$.

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We have 4 cases to discuss:

0. three edges on the new petal
1. two edges in the new petal
2. one edge in the new petal
3. no edges in the new petal, i.e., both in the existing F_k

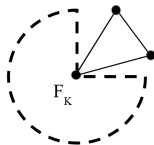


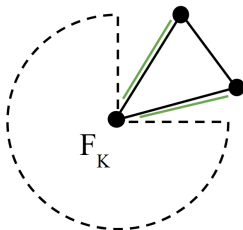
Figure 13: F_{k+1} built from F_k .

Theorem case 1 and 2

Proof.

For the first case, we want to show that the subgraph is either an edge blanket or has an internal edge. If it already has an internal edge then it doesn't really matter. But if it's an edge blanket, putting the two new edges on the new petal completes the edge blanket again.

In fact, this is also true for the second case because one edge is needed to touch the rest. Meaning we just proved two cases already.



Theorem case 3

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If the subgraph of F_k has no internal edges, then each petal should have a maximum edge count of 2 in the subgraph, resembling of a star graph. Meaning, we want to avoid 3 edges on each petal in the subgraph, and so our maximum edge count for F_k should be $2k$ without any internal edges.

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So, when we place $2k + 1$ edges in F_k 's subgraph, by a counting argument, one petal has to have 3 edges, creating an internal edge.

Theorem case 3 example

Proof.

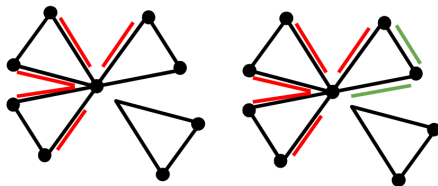


Figure 14: A petal being added onto a F_3 graph. Shows that 2 new edges in subgraph of F_3 creates an internal edge.



Cycle graphs

Definition

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- Whenever vertex count increases by three, edge density increases by two.
- At some points, edge densities are equal for two different vertex counts.

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If C_n is the cycle graph on n vertices, then the minimal edge density is:

$$\begin{cases} \frac{2}{3}n - 1, & n \equiv 0 \pmod{3} \\ \frac{2}{3}n - \frac{2}{3}, & n \equiv 1 \pmod{3} \\ \frac{2}{3}n - \frac{1}{3}, & n \equiv 2 \pmod{3}. \end{cases}$$

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We expect a proof similar to that of the flower graphs to work for our conjecture with cycle graphs:

- Use induction (three separate times) to prove C_n is d -edge dense for the appropriate d above.
- Prove that C_n is not $(d - 1)$ -edge dense for the appropriate d above.

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- Use induction (three separate times) to prove C_n is d -edge dense for the appropriate d above.
- Prove that C_n is not $(d - 1)$ -edge dense for the appropriate d above.

We prove the second bullet point by constructing appropriate counterexample subgraphs.

A partial solution

Proposition

If C_n is the cycle graph on n vertices, then the minimum possible edge density is at least:

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We will describe how to construct the relevant subgraph when $n \equiv 1 \pmod{3}$ [c.f. Figure 15].

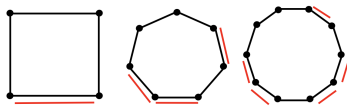


Figure 15: $n \equiv 1 \pmod{3}$ counterexamples for $(\frac{2}{3}n - \frac{2}{3} - 1)$ -edge dense.

vertices	4	7	10	13
edge density	2	4	6	8

What remains to be shown

Proving that all possible cycle graphs have the appropriate d -edge dense is much more difficult. We have to prove that, whenever you add three vertices to an existing graph, the d -edge density increases by two.

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Complete Graphs

One group of graphs we looked at are called complete graphs.

Definition

A **complete graph** with n vertices, denoted as K_n , is a graph such that every distinct pair of vertices share a unique edge.

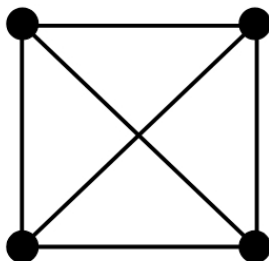


Figure 16: An example of a complete graph named K_4 .

Complete Graphs

Conjecture

Let $n \geq 3$. The minimum edge density for a complete graph K_n is $n - 2$.

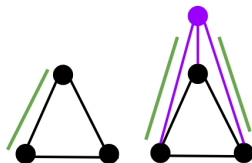


Figure 17: The graph K_3 becoming K_4 .

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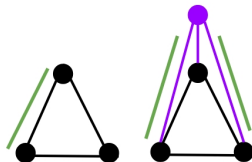


Figure 17: The graph K_3 becoming K_4 .

- We also used this approach to figure out K_6 was 4-edge dense by building off on the graph K_5 .

Remark

K_1 is a particularly special case. This is because K_1 , having one vertex, has no subgraphs with one edge which makes checking edge density trivial. Meaning K_1 can be any d -edge density.

Commutative algebra implications

Proposition

Let k be a field and $Q = k[x_1, \dots, x_n]$ be a polynomial ring in n variables over k . To each graph G with n vertices, one can associate an ideal $I_G \subseteq Q$. Call the quotient ring $R = Q/I_G$. When G is d -edge dense, then sometimes we obtain the following vanishing of Tor :

$$\mathrm{Tor}_1^Q(k, R) \cdot \mathrm{Tor}_d^Q(k, R) = 0.$$

Thank you!

Thank you all for taking the time to come see our presentation and we hope you have enjoyed it and found it intriguing, we know we did.



Acknowledgments

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- Eloísa Grifo & Mark Walker and our mentors, Isaiah Martinez and Sara Mueller
- UNL math department
- NSF RTG Grant DMS-2342256 Commutative Algebra at Nebraska for financial support.



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